IBM DATA SCIENCE PROFESSIONAL CERTIFICATE • CAPSTONE

SpaceX Falcon 9 First-Stage Landing Prediction

Predicting booster reusability through machine learning and interactive visual analytics.

AUTHOR

Data Science Candidate

GITHUB REPOSITORY

github.com/spacex-capstone-repo

PROJECT OVERVIEW

Executive Summary

Core Problem

Predicting launch reusability to determine actual flight cost structure.

Methodology

API ingestion, SQL analytics, Folium maps, and tuned classifiers.

Key Finding

83.33% classification accuracy across all optimized ML models.

Impact

$100M+ savings per commercial launch vs traditional rockets.

COMMERCIAL CONTEXT

Reusability Economics



SPACEX LAUNCH COST

$62M

Enabled by booster recovery

CONVENTIONAL LAUNCH COST

$165M+

Expendable stage architecture

DATA PIPELINE

Collection & Feature Engineering

1

SpaceX API Ingestion

- **Payloads:** Mass & orbit specs
- **Cores:** Reused counts & serials
- **Scope:** 90 Falcon 9 flights

2

Wikipedia Scraping

- **Soup parsing:** HTML table parsing
- **Targeting:** Launch site details
- **Validation:** Cross-referencing records

3

One-Hot Encoding

- **Transform:** Categorical expansion
- **Dimensions:** 83 total columns
- **Imputation:** Mean payload filling

EXPLORATORY ANALYSIS

Orbit & Payload Dynamics

ORBIT SUCCESS RATE

100% Reliability

ES-L1, GEO, HEO, and SSO missions

PAYLOAD THRESHOLD

2,000 – 6,000 kg

Optimal landing window margin

Key EDA Insights

- **Flight Progression:** Landing success improved rapidly after Flight 20.
- **Payload Mass:** Heavy payloads (>10k kg) restricted to drone ships.
- **GTO Challenges:** GTO orbits exhibit moderate 60% success rate.

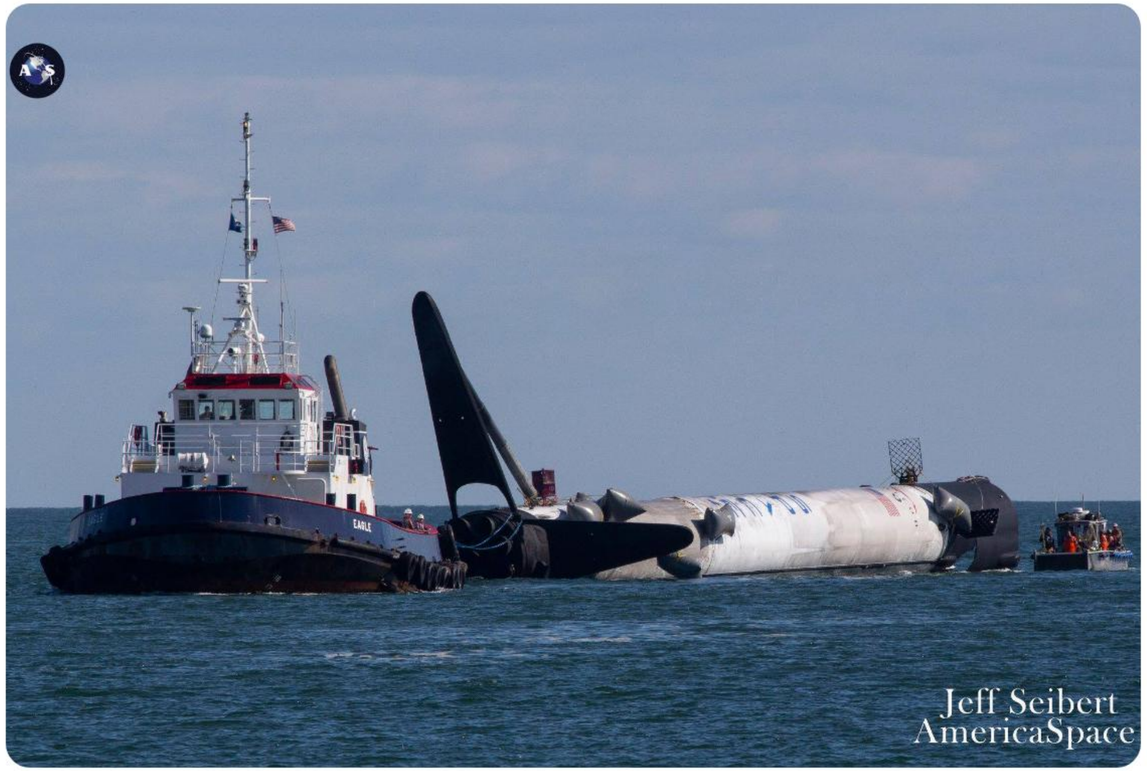
DATABASE QUERYING

SQL Key Insights

SQL Query Focus	Key Function	Aggregated Result
Unique Launch Sites	SELECT DISTINCT Launch_Site	**4 Primary Complexes**
Drone Ship Recoveries	COUNT(Landing_Outcome = 'True ASDS')	**41 Successful Landings**
Total Payload Carried	SUM(PAYLOAD_MASS__KG_)	**Combined Fleet Tonnage**
Max Payload Booster	WHERE Payload = MAX(Payload)	**15,600 kg Max Mass**

GEOSPATIAL INTELLIGENCE

Launch Site Proximity Analysis



Coastline Proximity

Minimizes risk over populated landmasses

Rail & Transport Links

Direct access for stage hardware transport

Urban Safety Buffer

Maintains safety distances from cities

DASHBOARDING

Plotly Dash Interactive App

Interactive Filters

Launch Site dropdown selector and Payload Range slider.

Site Success Ratios

KSC LC-39A accounts for 41.7% of total successful landings.

Payload Scatter Correlation

Color-coded booster version performance across mass ranges.

PREDICTIVE MODELING

Machine Learning Workflow

STEP 01

Train/Test Split

80% training set (N=72), 20% test set (N=18).

STEP 02

Standardization

StandardScaler normalization across all numeric features.

STEP 03

GridSearchCV

10-fold cross validation for hyperparameter tuning.

STEP 04

Evaluation

Validation scoring via confusion matrix analysis.

MODEL EVALUATION

Classification Performance Comparison

Model Algorithm	Best CV Score	Test Accuracy	Optimal Hyperparameters
Logistic Regression	84.64%	**83.33%**	C = 0.01, penalty = 'l2'
Support Vector Machine	84.82%	**83.33%**	kernel = 'sigmoid', C = 1.0
Decision Tree Classifier	87.50%	**83.33%**	criterion = 'gini', depth = 4
k-Nearest Neighbors	84.82%	**83.33%**	n_neighbors = 4, p = 1

STRATEGIC VALUE

Actionable Insights

Mass Optimization

Payloads under 5k kg maximize recovery probabilities.

Site Selection

KSC LC-39A offers highest operational reliability.

Model Choice

Logistic Regression recommended for simple deployment.

WRAP UP

Conclusion

PREDICTIVE ACCURACY ACHIEVED

83.33%

Reliable classification on unseen test dataset.

Data-Driven Bidding

Accurate landing predictions enable aggressive cost pricing.

Proven Scalability

Block 5 booster iterations demonstrate near-perfect recovery.

IBM CAPSTONE SUBMISSION

Thank You / Q&A

Full code, notebooks, and analysis available on GitHub.

github.com/spacex-capstone-repo

Image Sources



http://spaceflightnow.com/wp-content/uploads/2026/05/20260530_Starlink_17-41_liftoff_SpaceX-1.jpeg

Source: spaceflightnow.com



http://www.americaspace.com/wp-content/uploads/2018/12/47577655_2014408321984815_2300791062521708544_o.jpg

Source: www.americaspace.com